Algebraic substitution

Linear expressions

Substitute positive and negative values into linear expressions.

10 MINUTES / 8 QUESTIONS / AS-01

Name _____ Date _____

Show your working. Use brackets for negative values.

Starter: use the full set. Exit ticket: choose 2-3 questions.

1 Given $x = 8$

Evaluate $2x + 7$

2 Given $x = -4$

Evaluate $5x - 5$

3 Given $x = 6$

Evaluate $6x - 6$

4 Given $x = -3$

Evaluate $7x - 2$

5 Given $x = 4$

Evaluate $2x - 9$

6 Given $x = -3$

Evaluate $8x - 3$

7 Given $x = -4$

Evaluate $5x - 6$

8 Given $x = -2$

Evaluate $8x + 10$

Answer sheet

Linear expressions | 8 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 23

02 -25

03 30

04 -23

05 -1

06 -27

07 -26

08 -6

Worked solutions

Linear expressions

01 Given $x = 8$

$2x + 7$

Substitute $x = 8$.

$2\times 8 + 7$

$16 + 7$

23

02 Given $x = -4$

$5x - 5$

Substitute $x = -4$.

$5\times (-4) - 5$

$-20 - 5$

-25

03 Given $x = 6$

$6x - 6$

Substitute $x = 6$.

$6\times 6 - 6$

$36 - 6$

30

04 Given $x = -3$

$7x - 2$

Substitute $x = -3$.

$7\times (-3) - 2$

$-21 - 2$

-23

Worked solutions

Linear expressions

05 Given $x = 4$

$2x - 9$

Substitute $x = 4$.

$2\times 4 - 9$

$8 - 9$

-1

06 Given $x = -3$

$8x - 3$

Substitute $x = -3$.

$8\times (-3) - 3$

$-24 - 3$

-27

07 Given $x = -4$

$5x - 6$

Substitute $x = -4$.

$5\times (-4) - 6$

$-20 - 6$

-26

08 Given $x = -2$

$8x + 10$

Substitute $x = -2$.

$8\times (-2) + 10$

$-16 + 10$

-6